

Chapter 1

2024.2.22

Name:Chen Zhuoyan Student ID:523030910037

1. Name two advantages of digital data as compared to analog data.

Answer Area:

- Digital data can be stored, processed, and transmitted more efficiently.
- Digital circuits are less vulnerable to noise than analog circuits.

2. Name an analog quantity other than temperature and sound.

Answer Area:

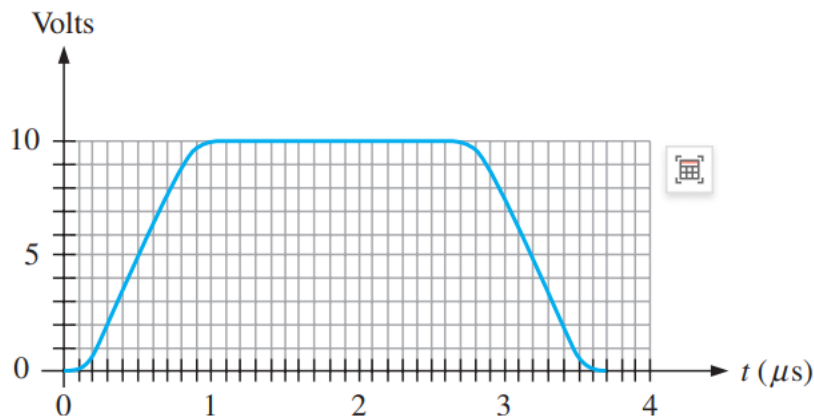
The height of a person.

3. List the sequence of levels (HIGH and LOW) that represent each of the following bit sequences
(b) 1 1 1 0 1 0 0 1

Answer Area:

HIGH HIGH HIGH LOW HIGH LOW LOW HIGH

4. For the pulse shown in Figure 1-20, graphically determine the following:
(a) rise time (b) fall time (c) pulse time (d) amplitude



Answer Area:

(a) $\therefore$ The time required for the pulse to go from its LOW level to its HIGH level is called the rise time.

$\therefore$ The rise time is $0.8 - 0.25 = 0.55 \mu s$

(b) $\therefore$ The time required for the pulse to go from its HIGH level to its LOW level is called the fall time.

$\therefore$ The fall time is $3.45 - 2.85 = 0.6 \mu s$

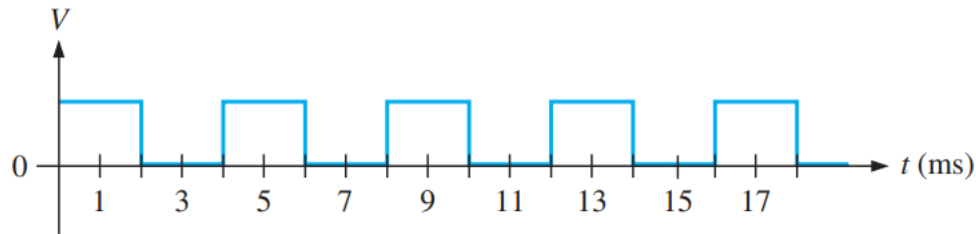
(c) $\therefore$ The pulse is often defined as the time interval between the 50 percent points on the

rising and falling edges.

$\therefore$ The pulse time is $3.2 - 0.5 = 2.7\mu s$.

(d) The amplitude is 10 volts.

5. Determine the period of the digital waveform in Figure 1-21

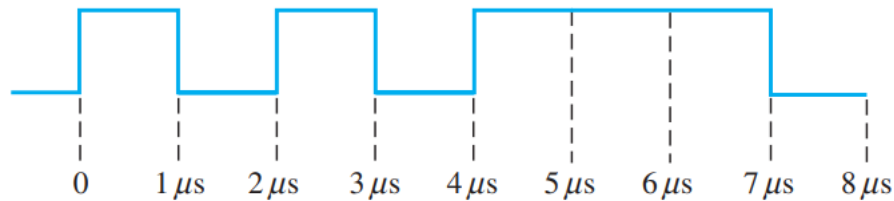


Answer area:

$\therefore$ The period is the fixed interval that the waveform repeats itself.

$\therefore$ According to the figure, we can know that the period is 4 ms.

6. Determine the bit sequence represented by the waveform in Figure 1-22. A bit time is $1\mu s$ in this case.



Answer Area:

$\therefore$ A bit is $1\mu s$ and HIGH = 1, LOW = 0

$\therefore$ The bit sequence is 10101110.

7. What is the total serial time for the eight bits in Figure 1-22? What is the total parallel transfer time?

Answer Area:

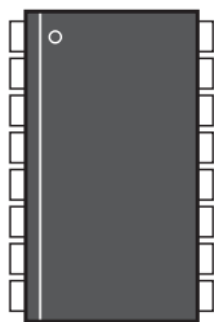
$\therefore$ When bits are transferred in serial form from one point to another, they are sent one bit at a time along a single conductor.

$\therefore$ It takes eight time intervals, that is $8\mu s$.

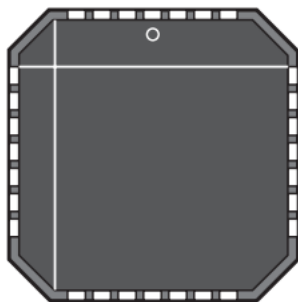
$\therefore$ When bits are transferred in parallel form, all the bits in a group are sent out on separate lines at the same time.

$\therefore$ It takes one time interval, that is $1\mu s$.

8. Label the pin numbers on the packages in Figure 1-23. Top views are shown.



(a)



(b)

Answer Area:

